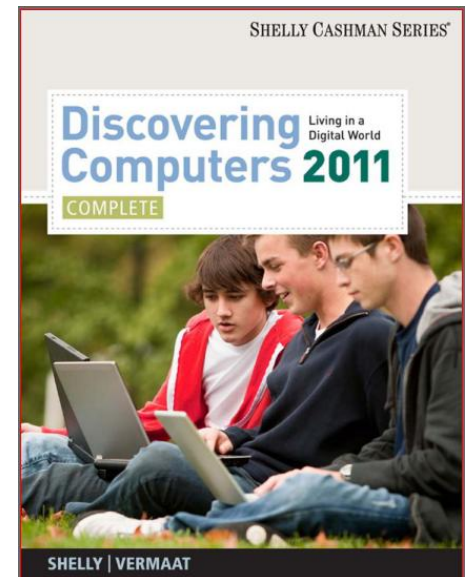


Lecture 5:

Artificial Intelligence



What is Artificial Intelligence?

- *In the simplest form, **Artificial Intelligence** means creating computing systems that have the ability to perform tasks like **humans** or even better than humans.*
- *It is a field of Computer Science that develops machines or software having human intelligence such as **speech recognition, decision making, and translation abilities.***

Applied Artificial Intelligence?

- *Applied Artificial Intelligence is the utilization of existing tools to extract useful information.*
- *The **Open-source technology** has a major role to play here for offering **free, easy-to-use** tools.*
- *However, it is important to note that tools are not a **magic box**.*
- *To utilize these tools effectively, you should know the potential biases and limitation of various models, and techniques.*

Research Artificial Intelligence?

- *Research Artificial Intelligence are a lot harder than Applied AI.*
- *This is due to the fact that **Research AI Open-source technology** requires you to understand **math, algorithm**, and several other related disciplines.*
- *For instance, how neural networks enhance the results of visual recognition.*

Types of Artificial Intelligence

- Artificial Intelligence has various subdivisions and other technologies such as Neural networks, machine learning, etc., but all these can be divided into three basic categories:
 1. Weak Artificial Intelligence
 2. True Artificial Intelligence
 3. Artificial SuperIntelligence

1. Weak Artificial Intelligence?

- ✓ Many artificial intelligence Experts points also define it as **narrow AI**.
- ✓ It is called weak because it doesn't have any intelligence or **self-awareness**.
- ✓ For instance, an assistant developed using narrow AI would have the *capability* of resolving your queries, but this assistant won't be able to have a *conversation* with you.
- ✓ They may have a **limited perspective** and **learning involved** in this perspective.



2. True Artificial Intelligence?

- ✓ True AI is also referred to as **strong AI**.
- ✓ The machines built using this technology would **react similarly to human beings**.
- ✓ The tasks that can be accomplished by human beings can also be achieved by these machines.
- ✓ However, there is still a lot of research going on in this area that is yet to show results.



3. Artificial Superintelligence?

- ✓ Artificial superintelligence is what makes us think that human extinction can be led by machines.
- ✓ Many scientists believe that this field has the power to create **machines** that can be **better** than **humans** in every field including **creativity** and **intellect**.



Importance of AI...

- ✓ AI is important in almost every field
- ✓ Everything from self-driving cars to smartwatches, a lot of aspects in our day-to-day lives utilize the power of AI.
- ✓ AI automates various repetitive tasks and learning. This technology has the ability to perform computerized, high-volume, and frequent tasks without human guidance.
- ✓ AI imparts intelligence to various products. For instance, new smartphones that come with Google Assistant.



Importance of AI...

- ✓ AI generates a **progressive algorithm** that learns over time. For example, if AI recommendations are being used in an e-commerce website, it means that this algorithm has the ability to predict what to recommend to users the next time.
- ✓ AI provides high-level accuracy which was impossible even a decade ago.

Think about it, how does your smartphone recognize your face?



Domains Included in Artificial Intelligence.

- ✓ Artificial intelligence is divided into **several domains** that work in specific fields.
- ✓ Here are a few of these domains:
 1. Robotics
 2. Neural Networks
 3. Fuzzy Logic System
 4. Expert Systems
 5. Natural Language Processing



1. Robotics

- ✓ Robotics is a branch of artificial intelligence that focuses on **creating artificial agents**.
- ✓ These robots **work** like humans and execute basic tasks such as **picking, serving, moving, destroying**, etc.
- ✓ For example, many restaurants now have robots to serve food quickly to the relevant tables.



2. Neural Networks

- ✓ Neural networks are really important in artificial intelligence. This domain forms algorithms to **detect patterns** and **trends**.
- ✓ The name is based on neurons of the human brain as the **structure** of the **neural networks** resembles the **human brain**.
- ✓ These systems learn with examples just as we learn from our mistakes or experiences.



3. Fuzzy Logic Systems

- ✓ Instead of explicitly stating true or false, a fuzzy logic system tells the degree of truth.
- ✓ These systems even have the capability to process noisy, imprecise, or distorted inputs.



4. Expert Systems

- ✓ Expert systems judge human decision-making capabilities.
- ✓ It is a machine that judges how an organization or a human being behaves or reacts according to their experience and knowledge in a given field.



5. Natural Language Processing (NLP)

- ✓ Natural language processing (NLP) is an essential field of artificial intelligence.
- ✓ It helps in various tasks such as **translation, sentiment analysis, named entity recognition, topic segmentation, and speech recognition.**



Applications Related to Artificial Intelligence.

- ✓ New applications of artificial intelligence emerge every day.
- ✓ There is no industry or field that is left untouched by this intelligent technology.
- ✓ Below are some of the leading applications of Artificial Intelligence.
 - i. Machine Translation
 - ii. Speech Recognition
 - iii. Chatbots
 - iv. Facial Recognition
 - v. Virtual Assistants



1. Machine Translation

- ✓ Machine translation is a field of artificial intelligence and **computational linguistics** that allow translation from one language to another.
- ✓ These systems additionally assist in **estimating turnaround time**, reviewing linguistics, analyzing files, managing language resource, etc.



2. Speech Recognition

- ✓ Similar to machine translation, speech recognition is a field of artificial intelligence that **recognizes** the **voice** and **converts** it to **text** in a computer system.
- ✓ Many effective speech recognition methods utilize long short-term memory system by deep learning.



3. Chatbots

- ✓ Chatbots are definitely one of the most useful and valuable inventions of artificial intelligence.
- ✓ These are **messaging bots** that respond to users and solve their queries.
- ✓ These bots have natural language processing capabilities that help them to analyze the question raised by the user and answer it appropriately.



4. Facial Recognition

- ✓ Facial recognition is an ability given to the machine to **recognize** the **facial pattern** of a human.
- ✓ These machines compare the facial features of the human to the ones given in the database.
- ✓ This artificial intelligence system is also referred to as **biometric artificial intelligence** as it analyzes the unique patterns of the face of an individual based on facial shape and texture.



5. Virtual Assistants

- ✓ Virtual assistants are often used synonymously to chatbots but they are not.
- ✓ Most of the virtual assistants now have the ability to interpret human voice and respond in a synthesized voice. For instance, Alexa.
- ✓ It is also the basic function of virtual assistants to interpret voice commands and execute them accordingly.



How To get Started?

✓ To get started with Artificial Intelligence, you need to first learn the basics of the following technologies.

1. SQL & Python
2. Machine Learning
3. Probability and Statistics



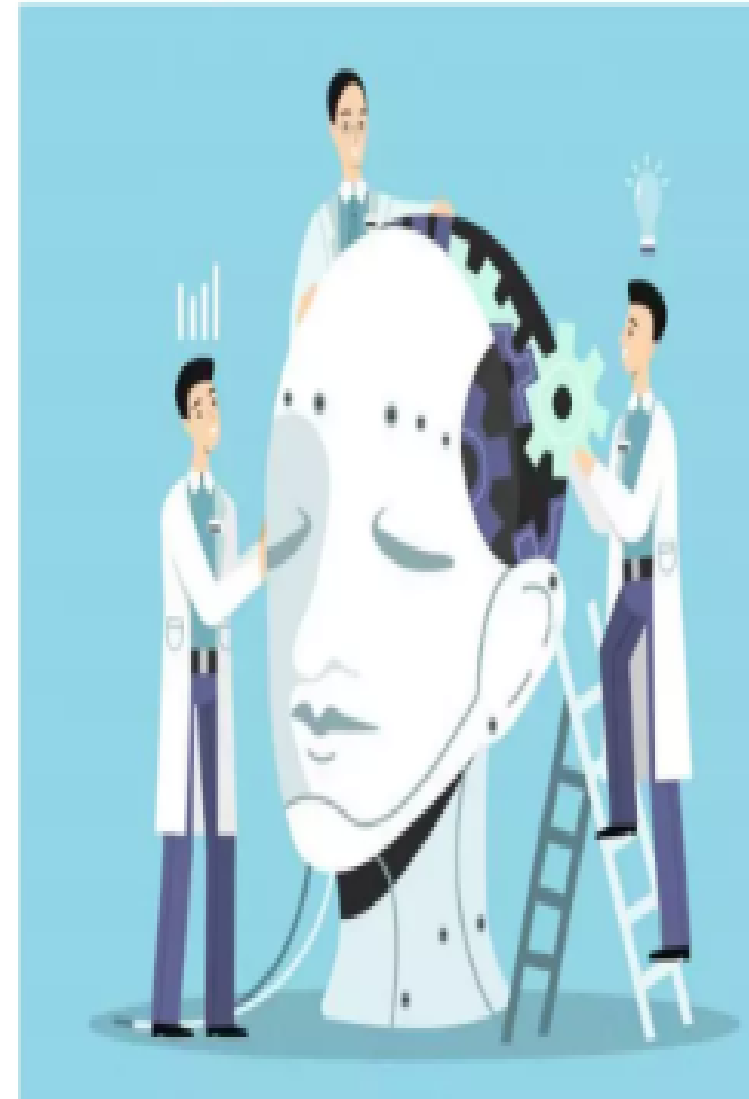
Learn SQL and Python.

- ✓ Python language allows you to maximize your *knowledge of hardware*. This language is used in collaboration with C and C++ and it delivers high-performance due to great libraries and toolkits.
- ✓ SQL is necessary for effective functioning. For instance, if your AI needs to work on updated data and most of your data is in a relational database, then you need to know SQL for correct execution.



Learn Machine Learning.

- ✓ Machine learning is an essential aspect of computer program development.
- ✓ It allows your systems to automatically learn from experiences and implement this knowledge for future enhancements.
- ✓ All this without human interference.



Learn Probability & Statistics.

- ✓ If you want to enter the field of machine learning, then probability theory and statistics are both the foundation of this domain.
- ✓ For deep understanding, you need to learn statistics and probability theory concepts such as Bayes' theorem.



Questions



End of segment